How light affects plants

Short science reading | one-page example | ages 9+ and adults

THE BIG IDEA

Plants need light to make food. During photosynthesis, green leaves use light energy, water and carbon dioxide to produce sugars; oxygen is released. Light matters, but more light does not always mean better growth: too much light or heat can also harm a plant.

1 A small, fair investigation

Three basil seedlings were placed in identical pots with the same soil. All were 4 cm tall on day 1. The only planned difference was their daily light exposure.

POT	LIGHT EXPOSURE	OTHER CONDITIONS
A	2 hours/day	same seed type, soil and water
B	6 hours/day	same seed type, soil and water
C	10 hours/day	same seed type, soil and water

2 Watering and timing

Each pot received exactly 50 mL of water on days 1, 3, 5, 7, 9 and 11. The final heights were recorded at the end of day 12. Look at the companion PNG to read those heights: they are not included in this PDF.

THINK LIKE A SCIENTIST

A pattern in one small illustration is not proof that light always causes taller plants. To make a stronger claim, repeat the test with more plants and record measurements over time.

3 Try it yourself

1. What was the changed variable? Name two things kept the same.
2. How much water did each plant receive across all six watering days?
3. Using the PNG, which pot finished tallest, and how many cm taller than A?
4. What additional evidence would make the conclusion more reliable?

This is invented sample data for testing Halo, not a real scientific experiment.